MPG No. 5.1.12 $2/m$ [Type I, monoclinic] [T tensor]

* Rank 0 tensor.

$$[T]$$

$$T = \mathbb{T}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} T_{xx} & 0 & T_{xz} \\ 0 & T_{yy} & 0 \\ T_{xz} & 0 & T_{zz} \end{bmatrix}$$

$$T_{xx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$T_{xz} = \mathbb{T}_{xz}^{(1)}$$
$$T_{yy} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$
$$T_{zz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ -T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = -\mathbb{M}_y^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & T_{xxzx} & 0 \\ T_{xxyy} & T_{yyyy} & T_{yyzz} & 0 & T_{yyzx} & 0 \\ T_{xxzz} & T_{yyzz} & T_{zzzz} & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & T_{yzyz} & 0 & T_{yzxy} \\ T_{xxzx} & T_{yyzx} & T_{zzzx} & 0 & T_{zxzx} & 0 \\ 0 & 0 & 0 & T_{yzxy} & 0 & T_{xyxy} \end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$
$$T_{xxyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)}$$
$$T_{xxzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + \mathbb{T}_v^{(1)}$$
$$T_{xxzx} = -\mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)}$$
$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$
$$T_{yyzz} = \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - \mathbb{T}_v^{(1)}$$
$$T_{yyzx} = 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)}$$
$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$
$$T_{zzzx} = \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)}$$
$$T_{yzyz} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - \mathbb{T}_v^{(2)}$$
$$T_{yzxy} = 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)}$$
$$T_{zxzx} = \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + \mathbb{T}_v^{(2)}$$
$$T_{xyxy} = \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & T_{xxyy} & T_{xxzz} & 0 & T_{xxzx} & 0 \\ -T_{xxyy} & 0 & T_{yyzz} & 0 & T_{yyzx} & 0 \\ -T_{xxzz} & -T_{yyzz} & 0 & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ -T_{xxzx} & -T_{yyzx} & -T_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} T_{xxyy} &= 4\mathbb{M}^{(1)}_{xyz} - 2\mathbb{T}^{(3)}_{v} \\ T_{xxzz} &= -4\mathbb{M}^{(1)}_{xyz} + 3\mathbb{T}^{(3)}_{u} - \mathbb{T}^{(3)}_{v} \\ T_{xxzx} &= -2\mathbb{M}^{(1)}_{y} + 2\mathbb{M}^{\alpha(1)}_{y} + 2\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz} \\ T_{yyzz} &= 4\mathbb{M}^{(1)}_{xyz} + 3\mathbb{T}^{(3)}_{u} + \mathbb{T}^{(3)}_{v} \\ T_{yyzx} &= -4\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz} \\ T_{zzzx} &= 2\mathbb{M}^{(1)}_{y} - 2\mathbb{M}^{\alpha(1)}_{y} + 2\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz} \\ T_{yzxy} &= \mathbb{M}^{(1)}_{y} + 4\mathbb{M}^{\alpha(1)}_{y} \end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} T_{yzyz} & 0 & T_{yzxy} \\ 0 & T_{zxzx} & 0 \\ T_{yzxy} & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned} T_{yzyz} &= \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} + 2\mathbb{T}^{(4)}_{v} \\ T_{yzxy} &= 2\mathbb{T}^{(4)}_{xz} \\ T_{zxzx} &= \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} - 2\mathbb{T}^{(4)}_{v} \\ T_{xyxy} &= \mathbb{T}^{(3)}_{0} + 4\mathbb{T}^{(4)}_{u} \end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{yzxy} = \mathbb{M}^{(2)}_{y}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxzx} & 0 \\ 0 & T_{yyzx} & 0 \\ 0 & T_{zzzx} & 0 \\ T_{yzyz} & 0 & T_{yzxy} \\ 0 & T_{zxzx} & 0 \\ T_{xyyz} & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned} T_{xxzx} &= \mathbb{M}^{(3)}_{y} - \mathbb{M}^{\alpha(2)}_{y} - \mathbb{M}^{\beta(2)}_{y} + 2\mathbb{T}^{(5)}_{xz} \\ T_{yyzx} &= \mathbb{M}^{(3)}_{y} + 2\mathbb{M}^{(4)}_{y} + 2\mathbb{M}^{\alpha(2)}_{y} \\ T_{zzzx} &= \mathbb{M}^{(3)}_{y} - \mathbb{M}^{\alpha(2)}_{y} + \mathbb{M}^{\beta(2)}_{y} - 2\mathbb{T}^{(5)}_{xz} \\ T_{yzyz} &= \mathbb{M}^{(2)}_{xyz} - 3\mathbb{T}^{(5)}_{u} - \mathbb{T}^{(5)}_{v} \\ T_{yzxy} &= \mathbb{M}^{(4)}_{y} - \mathbb{M}^{\alpha(2)}_{y} + \mathbb{M}^{\beta(2)}_{y} + \mathbb{T}^{(5)}_{xz} \\ T_{zxzx} &= \mathbb{M}^{(2)}_{xyz} + 3\mathbb{T}^{(5)}_{u} - \mathbb{T}^{(5)}_{v} \\ T_{xyyz} &= \mathbb{M}^{(4)}_{y} - \mathbb{M}^{\alpha(2)}_{y} - \mathbb{M}^{\beta(2)}_{y} - \mathbb{T}^{(5)}_{xz} \\ T_{xyxy} &= \mathbb{M}^{(2)}_{xyz} + 2\mathbb{T}^{(5)}_{v} \end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & T_{yzxy} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & T_{zxzx} & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & T_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
T_{yzyz} &= \mathbb{M}^{(3)}_{xyz} - 3\mathbb{T}^{(6)}_{u} - \mathbb{T}^{(6)}_{v} \\
T_{yzxy} &= \mathbb{M}^{(6)}_{y} - \mathbb{M}^{\alpha(3)}_{y} - \mathbb{M}^{\beta(3)}_{y} - \mathbb{T}^{(6)}_{xz} \\
T_{zxxx} &= \mathbb{M}^{(5)}_{y} - \mathbb{M}^{\alpha(3)}_{y} - \mathbb{M}^{\beta(3)}_{y} + 2\mathbb{T}^{(6)}_{xz} \\
T_{zxyy} &= \mathbb{M}^{(5)}_{y} + 2\mathbb{M}^{(6)}_{y} + 2\mathbb{M}^{\alpha(3)}_{y} \\
T_{zxzz} &= \mathbb{M}^{(5)}_{y} - \mathbb{M}^{\alpha(3)}_{y} + \mathbb{M}^{\beta(3)}_{y} - 2\mathbb{T}^{(6)}_{xz} \\
T_{zxzx} &= \mathbb{M}^{(3)}_{xyz} + 3\mathbb{T}^{(6)}_{u} - \mathbb{T}^{(6)}_{v} \\
T_{xyyz} &= \mathbb{M}^{(6)}_{y} - \mathbb{M}^{\alpha(3)}_{y} + \mathbb{M}^{\beta(3)}_{y} + \mathbb{T}^{(6)}_{xz} \\
T_{xyxy} &= \mathbb{M}^{(3)}_{xyz} + 2\mathbb{T}^{(6)}_{v}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
T_{xxxx} & T_{xxyy} & T_{xxzz} & 0 & T_{xxzx} & 0 & 0 & T_{xxxz} & 0 \\
T_{yyxx} & T_{yyyy} & T_{yyzz} & 0 & T_{yyzx} & 0 & 0 & T_{yyxz} & 0 \\
T_{zzxx} & T_{zzyy} & T_{zzzz} & 0 & T_{zzzx} & 0 & 0 & T_{zzxz} & 0 \\
0 & 0 & 0 & T_{yzyz} & 0 & T_{yzxy} & T_{yzzy} & 0 & T_{yzyx} \\
T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & T_{zxzx} & 0 & 0 & T_{zxxz} & 0 \\
0 & 0 & 0 & T_{xyyz} & 0 & T_{xyxy} & T_{xyzy} & 0 & T_{xyyx}
\end{bmatrix}$$

$$T_{xxxx} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(1)} + 4\mathbb{T}_v^{(2)}$$

$$T_{xxyy} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 2\mathbb{T}_v^{(3)}$$

$$T_{xxzz} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$

$$T_{xxzx} = -2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)}$$

$$T_{xxxz} = -2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)}$$

$$T_{yyxx} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} + 2\mathbb{T}_v^{(3)}$$

$$T_{yyyy} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} + 2\mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(1)} - 4\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(1)} - 4\mathbb{T}_v^{(2)}$$

$$T_{yyzz} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} + 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{yyzx} = \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{yyxz} = -\mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{(4)} - 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{zzxx} = 4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} + \mathbb{T}_v^{(1)} + \mathbb{T}_v^{(3)}$$

$$T_{zzyy} = -4\mathbb{M}_{xyz}^{(1)} + \mathbb{T}_0^{(1)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(1)} - 3\mathbb{T}_u^{(3)} - \mathbb{T}_v^{(1)} - \mathbb{T}_v^{(3)}$$

$$T_{zzzz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} + 2\mathbb{T}_4^{(1)} + 4\mathbb{T}_u^{(1)} + 8\mathbb{T}_u^{(2)}$$

$$T_{zzzx} = 2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)}$$

$$T_{zzxz} = 2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)}$$

$$T_{yzyz} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$

$$T_{yzxy} = \mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)}$$

$$T_{yzzy} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} - \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} - \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{yzyx} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)}$$

$$T_{zxxx} = 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)}$$

$$T_{zxyy} = 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} - \mathbb{T}_{xz}^{(3)}$$

$$T_{zxzz} = -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)}$$

$$T_{zxzx} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} + 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} - \mathbb{T}_v^{(5)}$$

$$T_{zxxz} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} - \mathbb{T}_{4u}^{(1)} + \mathbb{T}_{4v}^{(1)} - \mathbb{T}_4^{(1)} + \mathbb{T}_u^{(2)} - 3\mathbb{T}_u^{(5)} + \mathbb{T}_v^{(2)} + \mathbb{T}_v^{(5)}$$

$$T_{xyyz} = -\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)}$$

$$T_{xyxy} = \mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} + 2\mathbb{T}_v^{(5)}$$

$$T_{xyzy} = -\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)}$$

$$T_{xyyx} = -\mathbb{M}_{xyz}^{(2)} + \mathbb{T}_0^{(2)} + 2\mathbb{T}_{4u}^{(1)} - \mathbb{T}_4^{(1)} - 2\mathbb{T}_u^{(2)} - 2\mathbb{T}_v^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & T_{yzyz} & 0 & T_{yzxy} & T_{yzzy} & 0 & T_{yzyx} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & T_{zxzx} & 0 & 0 & T_{zxxz} & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & T_{xyxy} & T_{xyzy} & 0 & T_{xyyx} \end{bmatrix}$$

$$
\begin{aligned}
T_{yzyz} &= \mathbb{M}^{(3)}_{xyz} + \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} - 3\mathbb{T}^{(6)}_{u} + 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{yzxy} &= \mathbb{M}^{(2)}_{y} + \mathbb{M}^{(6)}_{y} - \mathbb{M}^{\alpha(3)}_{y} - \mathbb{M}^{\beta(3)}_{y} + 2\mathbb{T}^{(4)}_{xz} - \mathbb{T}^{(6)}_{xz} \\
T_{yzzy} &= \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(3)}_{0} + 2\mathbb{T}^{(4)}_{u} - 3\mathbb{T}^{(6)}_{u} - 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{yzyx} &= -\mathbb{M}^{(2)}_{y} + \mathbb{M}^{(6)}_{y} - \mathbb{M}^{\alpha(3)}_{y} - \mathbb{M}^{\beta(3)}_{y} - 2\mathbb{T}^{(4)}_{xz} - \mathbb{T}^{(6)}_{xz} \\
T_{zxxx} &= \mathbb{M}^{(5)}_{y} - \mathbb{M}^{\alpha(3)}_{y} - \mathbb{M}^{\beta(3)}_{y} + 2\mathbb{T}^{(6)}_{xz} \\
T_{zxyy} &= \mathbb{M}^{(5)}_{y} + 2\mathbb{M}^{(6)}_{y} + 2\mathbb{M}^{\alpha(3)}_{y} \\
T_{zxzz} &= \mathbb{M}^{(5)}_{y} - \mathbb{M}^{\alpha(3)}_{y} + \mathbb{M}^{\beta(3)}_{y} - 2\mathbb{T}^{(6)}_{xz} \\
T_{zxzx} &= \mathbb{M}^{(3)}_{xyz} + \mathbb{T}^{(3)}_{0} - 2\mathbb{T}^{(4)}_{u} + 3\mathbb{T}^{(6)}_{u} - 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{zxxz} &= \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(3)}_{0} + 2\mathbb{T}^{(4)}_{u} + 3\mathbb{T}^{(6)}_{u} + 2\mathbb{T}^{(4)}_{v} - \mathbb{T}^{(6)}_{v} \\
T_{xyyz} &= -\mathbb{M}^{(2)}_{y} + \mathbb{M}^{(6)}_{y} - \mathbb{M}^{\alpha(3)}_{y} + \mathbb{M}^{\beta(3)}_{y} + 2\mathbb{T}^{(4)}_{xz} + \mathbb{T}^{(6)}_{xz} \\
T_{xyxy} &= \mathbb{M}^{(3)}_{xyz} + \mathbb{T}^{(3)}_{0} + 4\mathbb{T}^{(4)}_{u} + 2\mathbb{T}^{(6)}_{v} \\
T_{xyzy} &= \mathbb{M}^{(2)}_{y} + \mathbb{M}^{(6)}_{y} - \mathbb{M}^{\alpha(3)}_{y} + \mathbb{M}^{\beta(3)}_{y} - 2\mathbb{T}^{(4)}_{xz} + \mathbb{T}^{(6)}_{xz} \\
T_{xyyx} &= \mathbb{M}^{(3)}_{xyz} - \mathbb{T}^{(3)}_{0} - 4\mathbb{T}^{(4)}_{u} + 2\mathbb{T}^{(6)}_{v}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
T_{xxxx} & 0 & T_{xxxz} \\
0 & T_{yyyy} & 0 \\
T_{zzzx} & 0 & T_{zzzz} \\
T_{yyzx} & 0 & T_{yyzz} \\
T_{zzxx} & 0 & T_{zzxz} \\
0 & T_{xxyy} & 0 \\
0 & T_{yzzy} & 0 \\
T_{zxxx} & 0 & T_{zxxz} \\
T_{xxyy} + T_{yyzz} - T_{yzzy} - T_{zxxz} + T_{zzxx} & 0 & T_{xyyz} \\
0 & T_{xyyz} + \frac{T_{zzxz}}{2} - \frac{T_{zzzx}}{2} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxxx} &= 3\mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{4} - 2\mathbb{T}^{(1)}_{u} - 4\mathbb{T}^{(2)}_{u} + 2\mathbb{T}^{(1)}_{v} + 4\mathbb{T}^{(2)}_{v} \\
T_{xxxz} &= -\mathbb{T}^{\alpha(1)}_{4y} - \mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{xz} + \mathbb{T}^{(3)}_{xz} - 2\mathbb{T}^{(5)}_{xz} \\
T_{yyyy} &= 3\mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} + \mathbb{T}^{(1)}_{4v} + 2\mathbb{T}^{(1)}_{4} - 2\mathbb{T}^{(1)}_{u} - 4\mathbb{T}^{(2)}_{u} - 2\mathbb{T}^{(1)}_{v} - 4\mathbb{T}^{(2)}_{v} \\
T_{zzzx} &= \mathbb{T}^{\alpha(1)}_{4y} - \mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{xz} + \mathbb{T}^{(3)}_{xz} - 2\mathbb{T}^{(5)}_{xz} \\
T_{zzzz} &= 3\mathbb{T}^{(1)}_{0} + 2\mathbb{T}^{(1)}_{4u} + 2\mathbb{T}^{(1)}_{4} + 4\mathbb{T}^{(1)}_{u} + 8\mathbb{T}^{(2)}_{u} \\
T_{yyzx} &= 2\mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz} + \mathbb{T}^{(3)}_{xz} \\
T_{yyzz} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(1)}_{u} + 3\mathbb{T}^{(3)}_{u} - \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v} \\
T_{zzxx} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} + \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(1)}_{u} - 3\mathbb{T}^{(3)}_{u} + \mathbb{T}^{(1)}_{v} + \mathbb{T}^{(3)}_{v} \\
T_{zzxz} &= \mathbb{T}^{\alpha(1)}_{4y} - \mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{xz} + \mathbb{T}^{(3)}_{xz} + 2\mathbb{T}^{(5)}_{xz} \\
T_{xxyy} &= \mathbb{T}^{(1)}_{0} + 2\mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4} - 2\mathbb{T}^{(1)}_{u} - 2\mathbb{T}^{(3)}_{v} \\
T_{yzzy} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} - \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(2)}_{u} + 3\mathbb{T}^{(5)}_{u} - \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v} \\
T_{zxxx} &= -\mathbb{T}^{\alpha(1)}_{4y} - \mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{xz} - \mathbb{T}^{(3)}_{xz} \\
T_{zxxz} &= \mathbb{T}^{(1)}_{0} - \mathbb{T}^{(1)}_{4u} + \mathbb{T}^{(1)}_{4v} - \mathbb{T}^{(1)}_{4} + \mathbb{T}^{(2)}_{u} - 3\mathbb{T}^{(5)}_{u} + \mathbb{T}^{(2)}_{v} + \mathbb{T}^{(5)}_{v} \\
T_{xyyz} &= 2\mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(2)}_{xz} - \mathbb{T}^{(5)}_{xz}
\end{aligned}
$$